SECUREBANK FINANCIAL | SECUREMOBILE EXECUTIVE RISK DASHBOARD

Lab 3b • Prepared by Hilda Odein Joshua-Jack • Registration number: C11/26/CGRCE/17247 • Evaluation period: 3 years

Current annual expected loss	Residual annual expected loss	Annual risk reduction	Initial investment	Annual maintenance
$35.00m	**$8.04m**	**$26.96m (77.0%)**	**$700,000**	**$105,000**

Top risks and recommended control investments

Risk	SLE	Annual probability	Current ALE	Control recommendation	Initial cost	Residual ALE
Database injection	$125.00m	25%	$31.25m	WAF (75%)	$150,000	$7.81m
Session hijacking	$7.50m	40%	$3.00m	MFA enhancement (95%)	$200,000	$0.15m
API authentication bypass	$5.00m	15%	$0.75m	API Security Gateway (90%)	$350,000	$0.075m

Financial case and decision

Three-year model: $80.89m expected loss avoided; $1.015m total control cost (including maintenance); portfolio ROI ≈ 7,869.7%; simple payback ≈ 9.5 days. These are modelled exposure estimates, not guaranteed cash savings. Approve the three targeted controls subject to validation of the $250 per-record loss assumption, WAF traffic coverage, staged testing and quarterly effectiveness reporting.

Phased plan: Days 1–30: immediate containment, transaction/session monitoring and loss-assumption validation. Days 31–90: deploy API gateway, WAF and MFA/session controls with tuning and retesting. Days 91–365: continuous control monitoring, quarterly assurance, secure SDLC and annual risk refresh.

Key modelling caveat: supplied exploit probabilities have no stated time horizon. This dashboard interprets them as annual probabilities of one loss event, not incident-frequency counts. ALE = SLE × annual event probability. WAF effectiveness is assumed to apply to database-injection traffic traversing the WAF. No control benefit is double-counted.